

Boundary-Supported Weyl-Electric Attribution in a Uniform-Density Oblate Spheroid

Single-Body Residual Geometry and Multipole Compression Scales in the Newtonian Limit

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Abstract

Finite matter contributes a finite curvature budget inside a continuous gravitational field. The field does not end at the boundary of matter, but neither is its curvature bookkeeping featureless. In the continuum regime of general relativity and its Newtonian weak-field limit, the metric remains seamless while source attribution, compression, and handoff can be structured. This paper develops a controlled weak-field test of that idea for one extended source: a uniform-density oblate spheroid.

The Ricci-Weyl decomposition separates locally stress-energy-constrained curvature from trace-free Weyl curvature. In the Newtonian limit, the trace of the tidal tensor, $\text{tr}(T) = 4\pi G\rho$, carries the Ricci/matter-density channel, while the trace-free tidal tensor $E_{ij} = T_{ij} - \frac{1}{3}\delta_{ij}\text{tr}(T)$ gives the static weak-field proxy for the electric part of the Weyl tensor. The central diagnostic distinction is therefore between Ricci-side compression and Weyl-electric custody: the former records the local source account, while the latter records shape, anisotropy, and trace-free tidal expression.

For the uniform spheroid, the density-gradient support is concentrated at the body boundary. This permits an exact boundary-supported reconstruction of the interior tidal tensor and an attribution record for the trace-free tidal field. Prototype computations for the fiducial spheroid $a = 1, c = 0.5, \rho_0 = G = 1$ validate the boundary reconstruction to near machine precision on compact interior points. Symmetry partitions recover the expected parity plane. The $\theta_c = \pi/4$ polar-cap partition produces no interior equality surface for the Weyl-electric Frobenius readable, and is therefore a residual-foliation witness rather than a parity surface.

A scalar finite-source comparison applied to the same cap-versus-belt partition behaves differently: its interior parity-volume fractions closely shadow the underlying source partition, while the Weyl-electric ledger closes tensorially, $E^{(A)} + E^{(K)} = E^{(\text{total})}$, without producing the same equality geometry. This divergence is the main interpretive result. Scalar parity geometry appears to act as a compressed Ricci-side attribution readable, whereas Weyl-electric reconstruction preserves a higher-fidelity trace-free curvature attribution channel. Exterior multipole tests further show how resolved source detail becomes compressible into lower-order representations at declared tolerances.

The construction introduces no new field equation, force, source, membrane, observable, or modification of general relativity. It is a readability result inside a declared weak-field model. The present work is static: it supplies a curvature-attribution ledger for a solved scene, but it does not yet provide the missing dynamical mechanism by which Ricci-side source custody becomes Weyl-electric curvature expression. A static boundary reconstruction must not be mistaken for instantaneous field updating. In GR, changes in stress-energy, shape, or boundary structure can only be expressed outside the future light cone of the change. The causal transformation from

source-local Ricci bookkeeping to propagated trace-free geometry remains the handoff problem exposed, but not solved, here.

1 Introduction

A finite distribution of matter does not carve a wall into spacetime. It contributes curvature to a continuous gravitational geometry. In the continuum model used by general relativity, the field does not tear at the surface of a planet, star, grain, or particle. Yet this does not mean that finite-source curvature identity is featureless everywhere. The metric can remain seamless while curvature custody changes hands.

This is the tension studied here. Finite matter carries a finite stress-energy account. That account is expressed through curvature, but the expression is not exhausted by one scalar tail. Near a resolved source, detailed matter structure, boundary structure, and anisotropy may be readable. Farther away, the same source may be represented through fewer degrees of freedom: mass, quadrupole, higher multipoles, or a collective parent context. Between these descriptions lies the handoff problem: where, and under which readable, does source-local curvature custody become contextual or compressed curvature expression?

The Ricci-Weyl decomposition is one of the standard ways of separating local and nonlocal aspects of curvature in general relativity [2, 3]. The Ricci part is algebraically constrained by stress-energy through Einstein’s equations. The Weyl tensor is trace-free and carries tidal, radiative, and nonlocal gravitational structure. The usual phrase “free gravitational field” is useful, but it can hide a bookkeeping question. If a Weyl-readable structure has a definite value in a modeled scene, where is that structure attributionally supported?

This paper does not propose a new answer to the Einstein field equations. It does not modify the Ricci-Weyl decomposition. It develops a restricted weak-field attribution construction in which the electric-Weyl proxy of a single extended body can be reconstructed, partitioned, and compared. The point is not to change gravity. The point is to make finite-source bookkeeping visible inside a field that remains continuous.

The clean testbed is a uniform-density oblate spheroid. It is extended rather than pointlike, analytically controlled, and anisotropic enough to produce nonzero trace-free interior tidal structure. Its spherical limit provides a natural null case, since a uniform-density sphere has no interior trace-free tidal field.

In the Newtonian limit, the tidal tensor is

$$T_{ij} = \partial_i \partial_j \Phi, \quad (1)$$

where Φ is the Newtonian potential. Its trace-free part is

$$E_{ij} = T_{ij} - \frac{1}{3} \delta_{ij} \text{tr}(T). \quad (2)$$

In static weak-field settings, E_{ij} is the leading Newtonian proxy for the electric part of the Weyl tensor in the matter rest frame. This paper uses E_{ij} as the declared trace-free tidal diagnostic.

For a uniform-density sharp-boundary body,

$$\rho(x) = \rho_0 \chi_\Omega(x), \quad (3)$$

so the density gradient is distributionally supported at the boundary. This permits a boundary-supported attribution construction. Boundary support means the distributional support of the density-gradient analogue in the declared Newtonian representation. It does not mean that a

physical current flows on a material membrane, and it does not turn boundary elements into new physical sources.

The central claim is narrow but not merely technical: a uniform-density oblate spheroid permits a boundary-supported reconstruction of its interior tidal and trace-free tidal structure in the Newtonian limit. Partitioning that support produces lawful single-body attribution residuals. Symmetric partitions recover exact parity planes. Non-symmetric partitions generate residual-imbalance fields whose level sets are witnesses, not parity surfaces unless they are equality surfaces. Exterior multipole truncation separately defines compression-faithful radii for declared tolerances.

The guiding sentence is this: the field is seamless, but curvature custody is structured.

2 Scope and non-claims

This paper begins in the Newtonian weak-field limit. The trace-free tidal tensor is used as the leading static proxy for electric Weyl structure. The source is a uniform-density sharp-boundary oblate spheroid. The boundary-supported attribution construction depends on the fact that the density-gradient support of this model is concentrated at the boundary.

The paper does not modify general relativity. It does not introduce a new field equation, force, source term, or material boundary. It does not claim that Weyl curvature is locally determined by matter gradients alone. It does not treat boundary patches as physical sources. It does not claim that every residual level set is a parity surface. It does not claim that a compression radius is a physical boundary. It does not claim observability. It does not claim that the Newtonian surface-integral identity automatically generalizes to the full Weyl tensor.

The governing rule is: a boundary partition may define an attribution residual, but it does not create new sources. A residual level set may witness imbalance, but only a declared equality surface is a parity surface.

3 Ricci-Weyl decomposition and weak-field electric-Weyl proxy

In four spacetime dimensions, the Riemann tensor decomposes as

$$R_{\alpha\beta\gamma\delta} = C_{\alpha\beta\gamma\delta} + (g \wedge P)_{\alpha\beta\gamma\delta}, \quad (4)$$

where $C_{\alpha\beta\gamma\delta}$ is the Weyl tensor and $P_{\mu\nu}$ is the Schouten tensor,

$$P_{\mu\nu} = \frac{1}{2} \left(R_{\mu\nu} - \frac{1}{6} R g_{\mu\nu} \right). \quad (5)$$

The Ricci-trace part is locally stress-energy-constrained through Einstein's equations. The Weyl part is the trace-free curvature not algebraically fixed by local stress-energy at that point. This does not make Weyl arbitrary. It means Weyl belongs to the full modeled scene: source arrangement, boundary data, initial data, matter gradients, propagation, and global constraints.

In the static Newtonian limit, the relevant curvature-readable quantity is the trace-free tidal tensor,

$$E_{ij} = T_{ij} - \frac{1}{3} \delta_{ij} \text{tr}(T), \quad T_{ij} = \partial_i \partial_j \Phi. \quad (6)$$

Scalar readables such as $\|E\|_F$, signed projections, eigenvalues, or eigenvector alignments may be defined only after the tensor object is constructed. Tensor construction comes first; scalarization comes second. In general,

$$\|\text{Riemann}\| - \|\text{RicciTracePart}\| \neq \|\text{Weyl}\|. \quad (7)$$

A scalar residual between curvature summaries is not automatically the Weyl component. The tensor decomposition must precede scalar comparison.

4 Conceptual split: Ricci compression and Weyl-electric custody

In the weak-field model, the tidal tensor separates into a trace and a trace-free part:

$$T_{ij} = E_{ij} + \frac{1}{3}\delta_{ij}\text{tr}(T). \quad (8)$$

The trace obeys

$$\text{tr}(T) = \nabla^2\Phi = 4\pi G\rho. \quad (9)$$

This is the Newtonian Ricci-side channel. It records where the finite matter distribution is locally registered as curvature.

The trace-free part,

$$E_{ij} = T_{ij} - \frac{1}{3}\delta_{ij}\text{tr}(T), \quad (10)$$

is the Weyl-electric channel in the static weak-field approximation. It records anisotropic tidal structure, shape information, and trace-free curvature expression. The two channels answer different bookkeeping questions. The Ricci-side channel asks where mass-energy is locally accounted for as curvature. The Weyl-electric channel asks how the finite source's shape and anisotropy are expressed as trace-free tidal geometry.

The word compression is therefore used in two distinct senses. Ricci compression means finite stress-energy represented as a local curvature trace account. Weyl or multipole compression means resolved source geometry simplified into exterior trace-free curvature structure, lower-order moments, or collective context. These are not separate forces and not separate worlds. They are different readable channels inside one curvature ledger.

This distinction prevents a common confusion. A scalar parity surface may express a compressed matter/Ricci-side bookkeeping relation, while a Weyl-electric attribution residual may refuse to produce the same surface because it preserves tensor direction and trace-free tidal structure. Agreement under symmetry is expected. Divergence under asymmetric partitioning is the point at which the richer geometry begins to speak.

5 Bianchi identity and density-gradient support

The contracted second Bianchi identity gives a differential bridge between Ricci structure and Weyl structure:

$$\nabla^\mu C_{\mu\nu\rho\sigma} = 2\nabla_{[\rho}P_{\sigma]\nu}. \quad (11)$$

Define

$$J_{\nu\rho\sigma} = 2\nabla_{[\rho}P_{\sigma]\nu}. \quad (12)$$

This quantity may be called a Bianchi current or Cotton-current in a chain-of-custody sense. It is not a new physical current. It is a standard differential expression built from Ricci/Schouten gradients. Through Einstein's equations it can be related to stress-energy gradients, with the usual matter-model and convention caveats.

The chain of custody is therefore

$$T_{\mu\nu} \rightarrow R_{\mu\nu} \rightarrow P_{\mu\nu} \rightarrow J_{\nu\rho\sigma} \rightarrow \nabla \cdot C. \quad (13)$$

The final arrow is deliberately $\nabla \cdot C$, not C . The contracted Bianchi identity gives a divergence constraint on Weyl, not a standalone evolution equation for all Weyl degrees of freedom.

Accordingly, this paper does not claim that Weyl is locally sourced by matter gradients in the same sense that a retarded electromagnetic potential is sourced by a current. The analogy is chain-of-custody only: Ricci/Schouten gradients constrain the divergence of Weyl, while homogeneous Weyl data, boundary conditions, initial data, gauge, and global geometry remain part of the full solution.

A schematic Green-function expression would require additional structure:

$$C(x) \sim \int G(x, y) J(y) dV_y + C_{\text{hom}}(x) + C_{\text{boundary}}(x). \quad (14)$$

The present paper does not attempt full covariant reconstruction. It uses the Bianchi-current idea only as a motivation for a weak-field attribution construction.

The Newtonian analogue is direct. With

$$\nabla^2 \Phi = 4\pi G \rho \quad (15)$$

and

$$E_{ij} = \partial_i \partial_j \Phi - \frac{1}{3} \delta_{ij} \nabla^2 \Phi, \quad (16)$$

one obtains

$$\partial^j E_{ij} = \frac{8\pi G}{3} \partial_i \rho. \quad (17)$$

For $\rho(x) = \rho_0 \chi_\Omega(x)$, the density gradient is

$$\nabla \rho = -\rho_0 \hat{n} \delta_{\partial\Omega}. \quad (18)$$

Thus, for a uniform-density sharp-boundary body, the Newtonian density-gradient support is concentrated at the boundary.

6 Uniform-density oblate spheroid model

Let

$$\Omega = \left\{ (x, y, z) : \frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} \leq 1 \right\}, \quad 0 < c \leq a, \quad (19)$$

with

$$\rho(x) = \rho_0 \chi_\Omega(x). \quad (20)$$

The eccentricity is

$$e = \sqrt{1 - \frac{c^2}{a^2}}. \quad (21)$$

The body is centered at the origin with symmetry axis z . The external context is asymptotically weak-field and otherwise empty. The only physical source is the spheroid.

For a uniform-density ellipsoid, the interior potential is quadratic. With the convention $\nabla^2 \Phi = 4\pi G \rho$ and $T_{ij} = \partial_i \partial_j \Phi$, the interior tidal tensor of the oblate spheroid is [1]

$$T_{ij} = 2\pi G \rho_0 \text{diag}(A_1, A_1, A_3), \quad (22)$$

where the standard oblate ellipsoidal coefficients are

$$A_1 = \frac{\sqrt{1-e^2}}{e^3} \sin^{-1} e - \frac{1-e^2}{e^2}, \quad (23)$$

and

$$A_3 = \frac{2}{e^2} - \frac{2\sqrt{1-e^2}}{e^3} \sin^{-1} e. \quad (24)$$

They satisfy $2A_1 + A_3 = 2$, so

$$\text{tr}(T) = 2\pi G\rho_0(2A_1 + A_3) = 4\pi G\rho_0, \quad (25)$$

as required by Poisson's equation.

In the spherical limit,

$$\lim_{e \rightarrow 0} A_1 = \lim_{e \rightarrow 0} A_3 = \frac{2}{3}. \quad (26)$$

The trace-free tidal tensor is

$$E_{ij} = T_{ij} - \frac{1}{3}\delta_{ij} \text{tr}(T). \quad (27)$$

For $0 < e < 1$, this may be written as

$$E_{ij} = 2\pi G\rho_0 \left(\frac{2}{3} - A_1 \right) \text{diag}(-1, -1, 2). \quad (28)$$

Thus $E_{xx} = E_{yy} < 0$ and $E_{zz} > 0$ for an oblate spheroid. The Frobenius norm is

$$\|E\|_F = 2\pi G\rho_0 \sqrt{6} \left(\frac{2}{3} - A_1 \right). \quad (29)$$

This gives the analytic scaling target for the trace-free tidal diagnostic. A successful reconstruction must recover the Poisson trace, recover the spatially uniform analytic tensor, recover zero trace-free tidal structure in the spherical limit, and recover nonzero axisymmetric trace-free structure for $e > 0$.

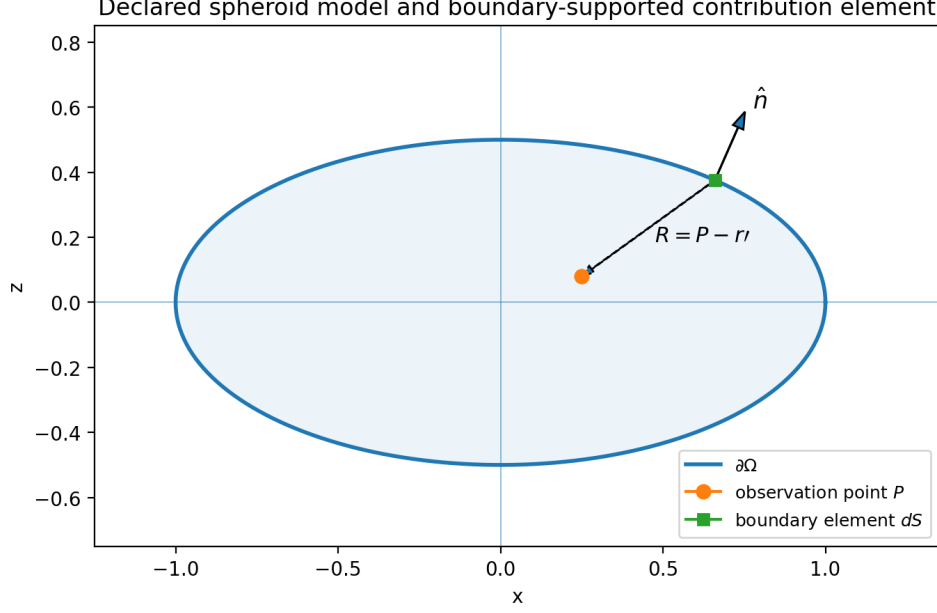


Figure 1: Declared spheroid model and boundary-supported contribution element in a meridional slice. The spheroid is the only physical source. Boundary elements represent derived density-gradient support in the Newtonian attribution construction, not additional sources or a material membrane.

7 Boundary-supported reconstruction of the interior tidal tensor

Use the convention

$$\Phi(P) = -G\rho_0 \int_{\Omega} \frac{1}{|P - r'|} dV', \quad (30)$$

so that $\nabla^2 \Phi = 4\pi G\rho$ and $g_i = -\partial_i \Phi$.

Let P be an interior observation point and let $r' \in \partial\Omega$. Define

$$R = P - r', \quad R = |P - r'|. \quad (31)$$

Set

$$I(P) = \int_{\Omega} \frac{1}{|P - r'|} dV'. \quad (32)$$

Since

$$\partial_{P_j} \left(\frac{1}{R} \right) = -\frac{R_j}{R^3}, \quad \partial_{r'_j} \left(\frac{1}{R} \right) = \frac{R_j}{R^3}, \quad (33)$$

we have $\partial_{P_j}(1/R) = -\partial_{r'_j}(1/R)$. Therefore,

$$\partial_{P_j} I(P) = - \int_{\Omega} \partial_{r'_j} \left(\frac{1}{|P - r'|} \right) dV'. \quad (34)$$

By the divergence theorem,

$$\partial_{P_j} I(P) = - \int_{\partial\Omega} \hat{n}_j(r') \frac{1}{R} dS. \quad (35)$$

Differentiating once more,

$$\partial_{P_i} \partial_{P_j} I(P) = \int_{\partial\Omega} \hat{n}_j(r') \frac{R_i}{R^3} dS. \quad (36)$$

Because $\Phi(P) = -G\rho_0 I(P)$, the tidal tensor is

$$T_{ij}(P) = \partial_{P_i} \partial_{P_j} \Phi(P) = -G\rho_0 \int_{\partial\Omega} \hat{n}_j(r') \frac{R_i}{R^3} dS. \quad (37)$$

For numerical work, it is useful to use the symmetrized form

$$T_{ij}(P) = -\frac{G\rho_0}{2} \int_{\partial\Omega} \frac{\hat{n}_i(r') R_j + \hat{n}_j(r') R_i}{R^3} dS. \quad (38)$$

The trace gives

$$T^i_i(P) = -G\rho_0 \int_{\partial\Omega} \frac{\hat{n}(r') \cdot (P - r')}{|P - r'|^3} dS. \quad (39)$$

For $P \in \Omega$,

$$\int_{\partial\Omega} \frac{\hat{n}(r') \cdot (P - r')}{|P - r'|^3} dS = -4\pi, \quad (40)$$

so $T^i_i(P) = 4\pi G\rho_0$, as required.

For a uniform sphere of radius a and an observation point at the center, $r' = a\hat{s}$, $R = -a\hat{s}$, and $\hat{n} = \hat{s}$. Then

$$\hat{n}_j \frac{R_i}{R^3} dS = -\hat{s}_i \hat{s}_j d\Omega. \quad (41)$$

Thus,

$$T_{ij} = G\rho_0 \int \hat{s}_i \hat{s}_j d\Omega = \frac{4\pi G\rho_0}{3} \delta_{ij}, \quad (42)$$

and the trace-free tensor vanishes.

The trace-free tidal diagnostic is then

$$E_{ij}(P) = T_{ij}(P) - \frac{1}{3} \delta_{ij} T^k_k(P). \quad (43)$$

A direct second-derivative volume derivation must keep the distributional contact term of $\partial_i \partial_j (1/R)$. The boundary derivation above avoids losing that term by applying the divergence theorem after the first derivative. Classical potential theory provides the relevant background for such Newtonian boundary representations [4].

The boundary integrand defines a per-boundary-element contribution,

$$dT_{ij}(r' \rightarrow P) = -\frac{G\rho_0}{2} \frac{\hat{n}_i(r') R_j + \hat{n}_j(r') R_i}{R^3} dS, \quad (44)$$

and

$$dE_{ij}(r' \rightarrow P) = \text{TF}[dT_{ij}(r' \rightarrow P)]. \quad (45)$$

A Frobenius density may be defined as

$$d\mathcal{E}_F(r' \rightarrow P) = \|dE(r' \rightarrow P)\|_F. \quad (46)$$

A signed projection onto the analytic interior direction may be defined as

$$d\mathcal{E}_{\parallel}(r' \rightarrow P) = \langle dE(r' \rightarrow P), \hat{E}_{\text{int}} \rangle_F. \quad (47)$$

These maps are witnesses. The primary numerical object is the table of tensor contributions.

Near-boundary observation points require higher boundary resolution or local singularity-aware quadrature. The kernel becomes sharply peaked near the closest boundary point as $P \rightarrow \partial\Omega$. Validation should therefore distinguish compact interior points, which are comfortably separated from the boundary, from near-boundary stress tests.

8 Boundary attribution densities

The boundary-supported reconstruction allows one to ask how different regions of the body boundary contribute to the trace-free tidal field at a selected interior point. At each observation point P , the map

$$r' \mapsto \|dE(r' \rightarrow P)\|_F \quad (48)$$

gives a nonnegative attribution-density witness. It shows where the magnitude of the boundary-element contribution is largest. The signed projection

$$r' \mapsto \langle dE(r' \rightarrow P), \hat{E}_{\text{int}} \rangle_F \quad (49)$$

shows whether a boundary element reinforces or opposes the analytic interior trace-free pattern.

These attribution maps are not independent evidence. They are visualizations of the boundary-element contribution table. They must not be interpreted as new surface sources. Their role is to show how the declared reconstruction distributes contribution across the boundary support.

9 Single-body boundary partitions and residual fields

A boundary partition divides the derived support:

$$\partial\Omega = \partial\Omega_A \cup \partial\Omega_K. \quad (50)$$

The labels A and K name attribution jurisdictions, not separate bodies.

Each jurisdiction contributes

$$E_{ij}^{(A)}(P) = \int_{\partial\Omega_A} dE_{ij}(r' \rightarrow P), \quad (51)$$

and

$$E_{ij}^{(K)}(P) = \int_{\partial\Omega_K} dE_{ij}(r' \rightarrow P). \quad (52)$$

The total is $E_{ij}(P) = E_{ij}^{(A)}(P) + E_{ij}^{(K)}(P)$. For the Frobenius readable,

$$\Psi_F[A|K](P) = \|E^{(A)}(P)\|_F - \|E^{(K)}(P)\|_F. \quad (53)$$

The equality locus is

$$\Sigma_F[A|K] = \{P \in \Omega : \Psi_F[A|K](P) = 0\}. \quad (54)$$

If it exists and is well-posed, this is a single-body boundary-attribution parity surface. A zero set of a declared attribution residual may be a parity surface. A nonzero level set is not a parity surface. Therefore, residual foliations are not generalized parity surfaces. They are witnesses around a residual field.

10 Symmetry parity planes as validation tests

The first validation partitions are symmetry partitions. For northern versus southern boundary support,

$$\partial\Omega_A = \{z' > 0\}, \quad \partial\Omega_K = \{z' < 0\}. \quad (55)$$

By reflection symmetry, the parity surface is $z = 0$. For $x > 0$ versus $x < 0$,

$$\partial\Omega_A = \{x' > 0\}, \quad \partial\Omega_K = \{x' < 0\}. \quad (56)$$

By reflection symmetry, the parity surface is $x = 0$.

These are not discoveries. They are validation cases. The extraction must recover the expected planes without tuning. A symmetry validation fails if the zero contour is displaced from the expected plane without declared reason, breaks into numerical fragments, depends strongly on quadrature resolution after convergence should have been reached, violates the expected reflection symmetry, or is inferred from a figure rather than the residual field.

11 Asymmetric partitions and residual-foliation witnesses

For an asymmetric partition, such as polar caps versus belt, define

$$\partial\Omega_A = \{r' \in \partial\Omega : |\cos \theta'| \geq \cos \theta_c\}, \quad (57)$$

and

$$\partial\Omega_K = \partial\Omega \setminus \partial\Omega_A. \quad (58)$$

The residual $\Psi_F^{(\theta_c)}(P)$ may not have a zero set. If $\Psi_F^{(\theta_c)}(P) \neq 0$ for all $P \in \Omega$, then no parity surface exists. The level sets

$$\mathcal{L}_c = \{P \in \Omega : \Psi_F(P) = c\} \quad (59)$$

are residual-foliation witnesses. They may show stable attribution imbalance, but they are not parity surfaces unless $c = 0$.

This distinction is mandatory. Parity is special because it is equality. Nonzero level sets do not inherit the interpretation of parity. Useful sweeps include cap angle θ_c , eccentricity e , and diagnostic readable. A retained asymmetric witness must earn retention through stability, parameter coherence, readable comparison, compression linkage, or relation to independent tidal/eigenstructure witnesses.

12 Multipole compression-faithful radii

The exterior potential of an axisymmetric body may be expanded as [5, 6]

$$\Phi(r, \theta) = -G \sum_{\ell=0}^{\infty} \frac{q_\ell}{r^{\ell+1}} P_\ell(\cos \theta). \quad (60)$$

A truncation gives

$$\Phi_{L_{\max}}(r, \theta) = -G \sum_{\ell=0}^{L_{\max}} \frac{q_\ell}{r^{\ell+1}} P_\ell(\cos \theta). \quad (61)$$

Define an error, for example,

$$\epsilon_{L_{\max}}(r) = \max_{\theta} \frac{|\Phi_{L_{\max}}(r, \theta) - \Phi_{\text{ref}}(r, \theta)|}{|\Phi_{\text{ref}}(r, \theta)|}. \quad (62)$$

Given tolerance τ ,

$$r_\tau(L_{\max}) = \inf \{r : \epsilon_{L_{\max}}(r') \leq \tau \text{ for all } r' \geq r\}. \quad (63)$$

This is the compression-faithful radius.

A compression-faithful radius is an approximation-validity boundary, not a parity surface. It is not a parity surface by default because it is defined by an inequality, not an equality between a declared contribution and a declared context. It becomes part of a parity construction only if a separate equality rule is declared, such as omitted multipole contribution equaling environmental context or retained structure.

Beyond $r_\tau(L_{\max})$, the body may be represented by multipoles through $L_{\max}$ for the declared exterior task and tolerance. This does not erase interior information. It authorizes a declared approximation. Potential-compression, acceleration-compression, tidal-compression, and trace-free-tidal compression radii should be tracked separately. They need not coincide.

13 Prototype numerical validation

The preceding sections define the attribution construction. We now report a minimal prototype validation for the fiducial spheroid

$$a = 1, \quad c = 0.5, \quad \rho_0 = 1, \quad G = 1. \quad (64)$$

The eccentricity is $e \approx 0.8660254$. The oblate coefficients are

$$A_1 \approx 0.4727997174, \quad A_3 \approx 1.0544005651, \quad (65)$$

with $2A_1 + A_3 = 2$. The analytic interior tidal tensor is therefore

$$T_{\text{analytic}} \approx \text{diag}(2.97068824, 2.97068824, 6.62499414), \quad (66)$$

and the trace-free tidal tensor is

$$E_{\text{analytic}} \approx \text{diag}(-1.21810197, -1.21810197, 2.43620393). \quad (67)$$

The trace satisfies $\text{tr}(T) = 4\pi$.

13.1 Boundary reconstruction convergence

Using the symmetrized boundary formula, we evaluated the reconstruction at several compact-interior points. Boundary quadrature used Gauss-Legendre nodes in $u = \cos \theta$ and uniform sampling in ϕ .

The reconstruction converges rapidly to the analytic tensor. In the fiducial run, the maximum relative Frobenius error over tested interior points fell from approximately 1.6×10^{-4} at 288 boundary nodes to approximately 3×10^{-14} at 20,000 boundary nodes. The trace-free tensor showed the same convergence pattern, reaching relative errors of order 10^{-13} at high quadrature resolution. This validates the boundary-supported reconstruction on compact interior points away from the boundary.

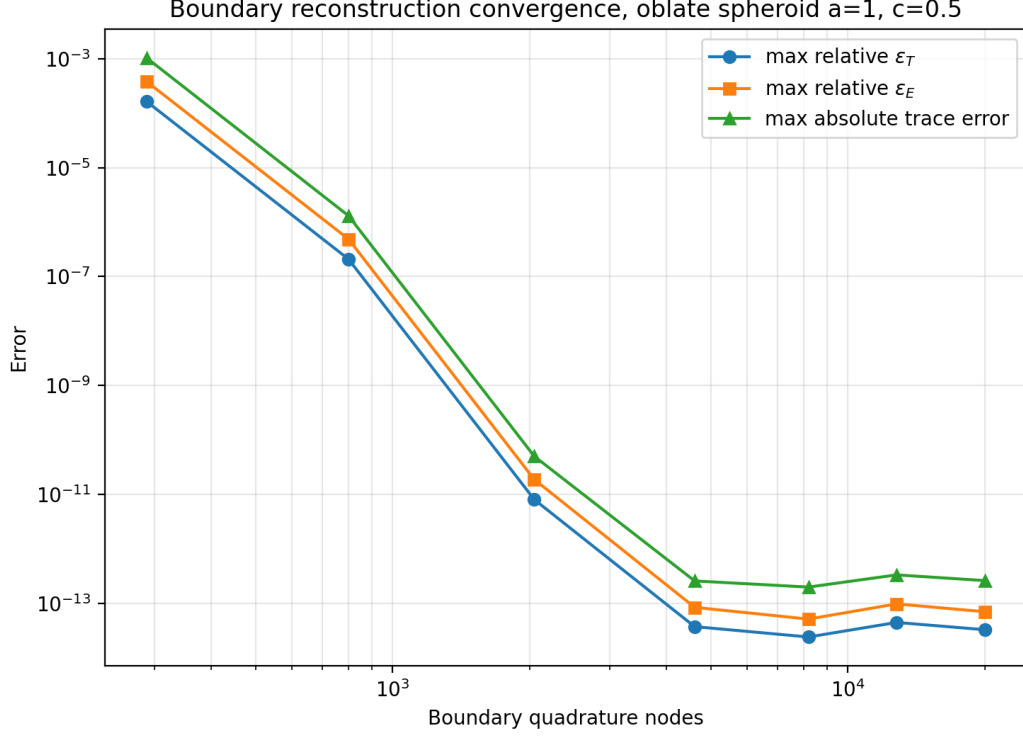


Figure 2: Boundary-supported reconstruction convergence for the fiducial oblate spheroid $a = 1, c = 0.5, \rho_0 = G = 1$. The symmetrized boundary integral recovers the analytic interior tidal tensor and trace-free tidal tensor to near machine precision for compact interior points. The trace error verifies recovery of $\text{tr}(T) = 4\pi G\rho_0$.

13.2 Boundary attribution maps

Boundary attribution maps were generated for selected interior observation points. The maps display $\log_{10} \|dE(r' \rightarrow P)\|_F$ and $\langle dE(r' \rightarrow P), \hat{E}_{\text{int}} \rangle_F$. At the center, the maps are axisymmetric and independent of ϕ , as expected. At an off-axis point, the maps lose azimuthal symmetry and show localized boundary regions contributing more strongly. These maps are witness maps only. They visualize the contribution table and do not constitute independent evidence or new source structure.

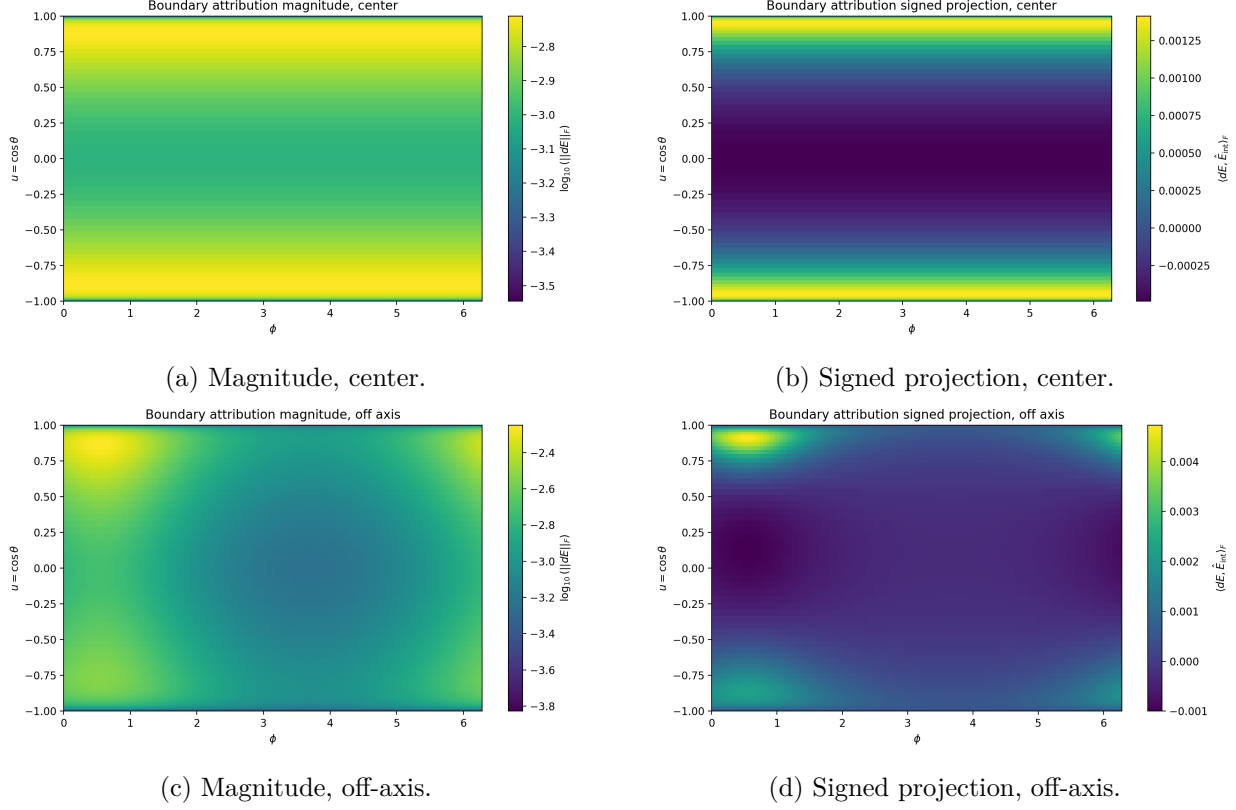


Figure 3: Boundary attribution density maps for selected interior points. Magnitude panels show $\log_{10} \|dE\|_F$; signed panels show projection onto the analytic interior trace-free tidal direction. The center point preserves axial symmetry, while an off-axis point produces localized attribution structure. These maps are visualizations of boundary-element tensor contributions, not new physical sources.

13.3 Symmetry partition validation

For the north-south boundary partition, $\partial\Omega_A = \{z' > 0\}$ and $\partial\Omega_K = \{z' < 0\}$, the Frobenius residual $\Psi_F(P) = \|E^{(A)}(P)\|_F - \|E^{(K)}(P)\|_F$ recovers the expected zero contour $z = 0$. The central symmetry plane is cleanly recovered. Small boundary-adjacent artifacts appear near the lateral edge of the plotted meridional slice. These are consistent with the expected near-boundary quadrature stress and are not interpreted as physical features.

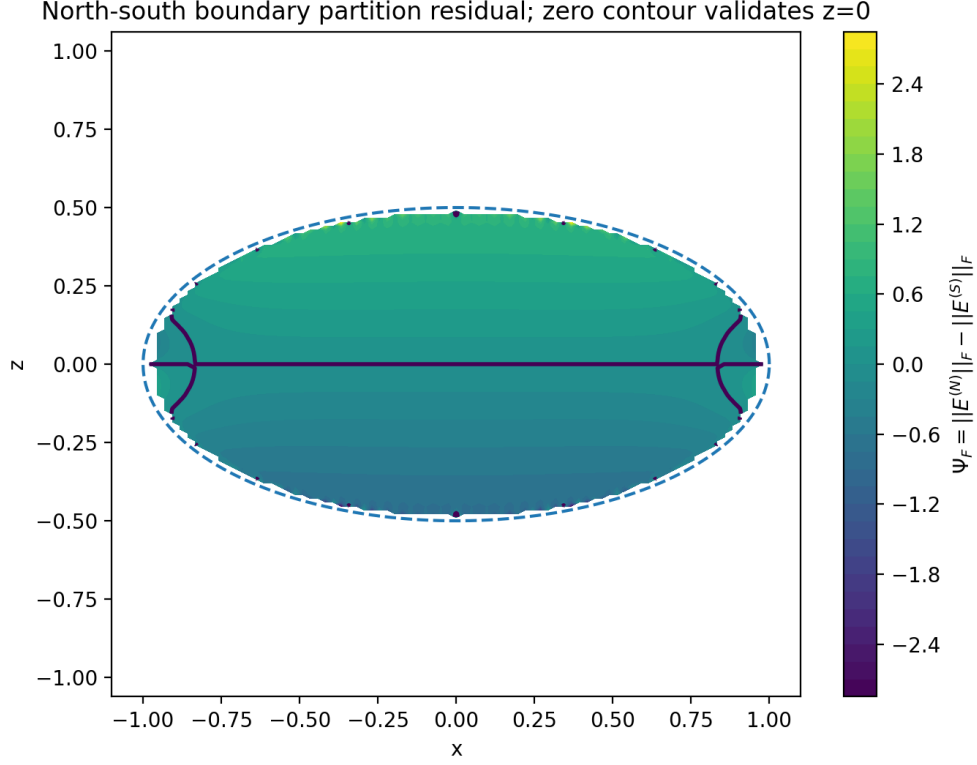


Figure 4: North-south boundary partition residual in a meridional slice. The zero contour recovers the expected equatorial parity plane $z = 0$. Boundary-adjacent artifacts near the lateral edge reflect near-boundary quadrature stress and are not interpreted as physical structure.

13.4 Asymmetric cap-versus-belt partition

For the polar-cap-versus-belt partition with $\theta_c = \pi/4$, the tested residual range was

$$\Psi_{\min} \approx 0.2136, \quad \Psi_{\max} \approx 2.9957. \quad (68)$$

Thus $0 \notin [\Psi_{\min}, \Psi_{\max}]$, and no interior equality surface exists for this partition and Frobenius readable. The resulting contours are therefore residual-foliation witnesses, not parity surfaces.

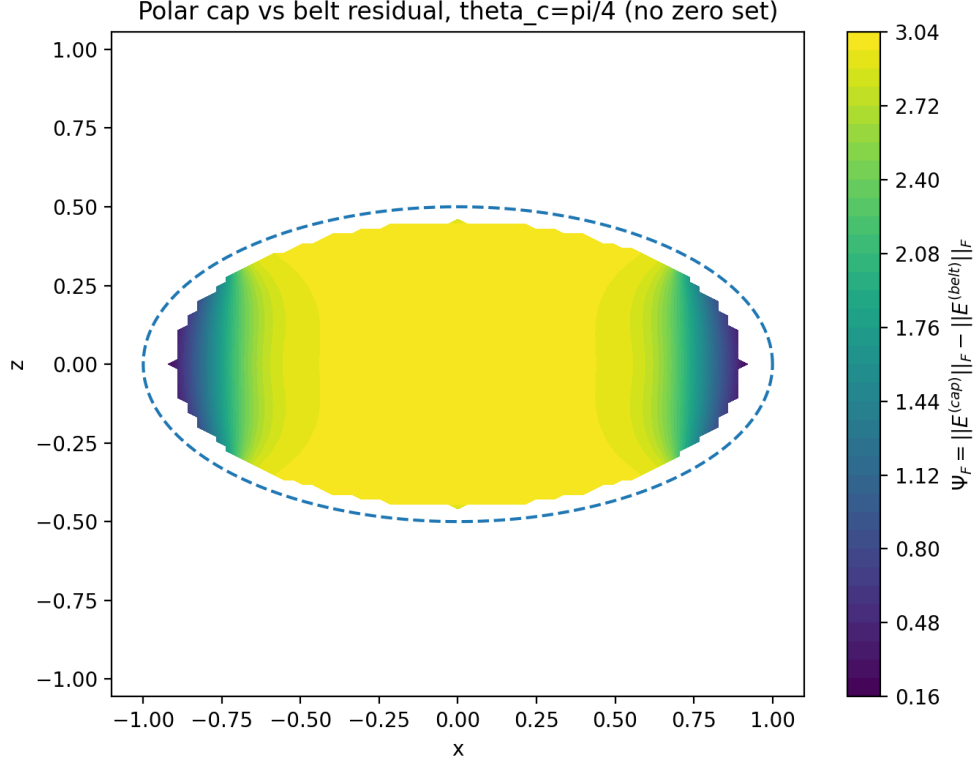


Figure 5: Polar-cap versus belt residual for $\theta_c = \pi/4$. The residual remains strictly positive over the sampled interior, with $\Psi_{\min} \approx 0.2136$. Therefore this partition does not produce an interior parity surface for the Frobenius readable. The displayed contours are residual-foliation witnesses.

13.5 Cap-angle sweep

A coarse cap-angle sweep shows that zero existence depends on the partition angle. Smaller cap angles produce residual ranges that cross zero, while larger cap angles become strictly one-sided. In the prototype sweep, the transition occurred between approximately $\theta_c \approx 0.67$ and $\theta_c \approx 0.74$ radians. This transition should be treated as exploratory until refined with higher resolution and contour tracking. Nevertheless, it is an important result: the asymmetric partition family contains both parity-producing and non-parity regimes.

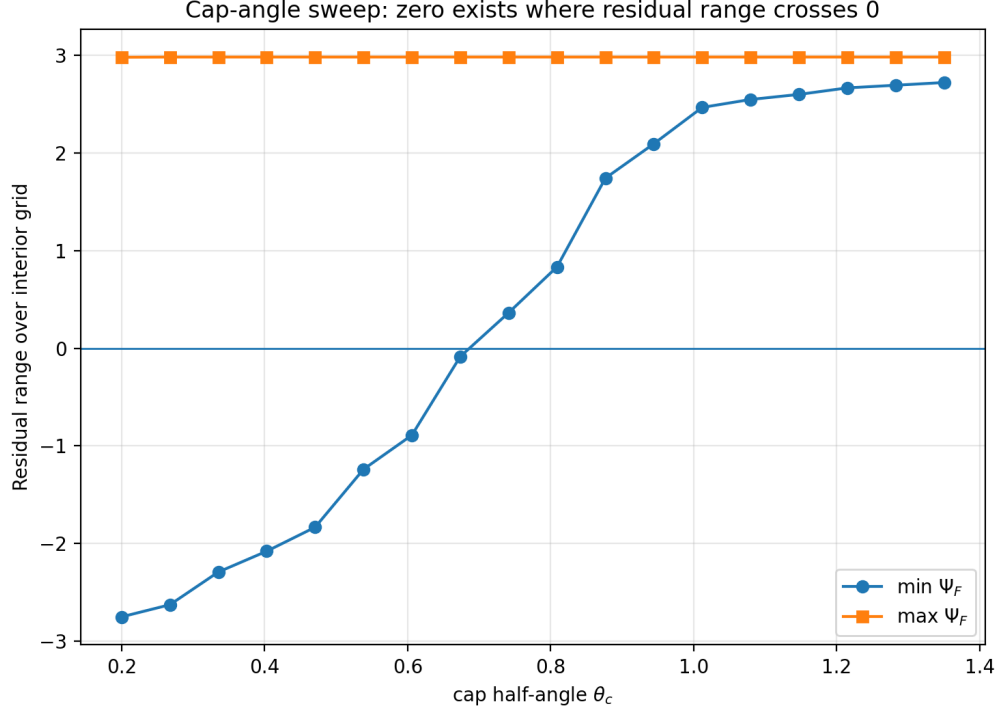


Figure 6: Cap-angle sweep for the polar-cap-versus-belt partition family. A zero set exists when the residual range crosses zero. The prototype scan shows a transition from parity-producing partitions to strictly one-sided residual-witness partitions between $\theta_c \approx 0.67$ and $\theta_c \approx 0.74$ radians. This sweep is exploratory and should be refined before stronger claims are made.

13.6 Eccentricity scaling

The analytic trace-free tidal strength is

$$\|E\|_F = 2\pi G \rho_0 \sqrt{6} \left(\frac{2}{3} - A_1(e) \right). \quad (69)$$

The prototype plot confirms the expected behavior: the trace-free tidal strength vanishes in the spherical limit and grows with oblateness.

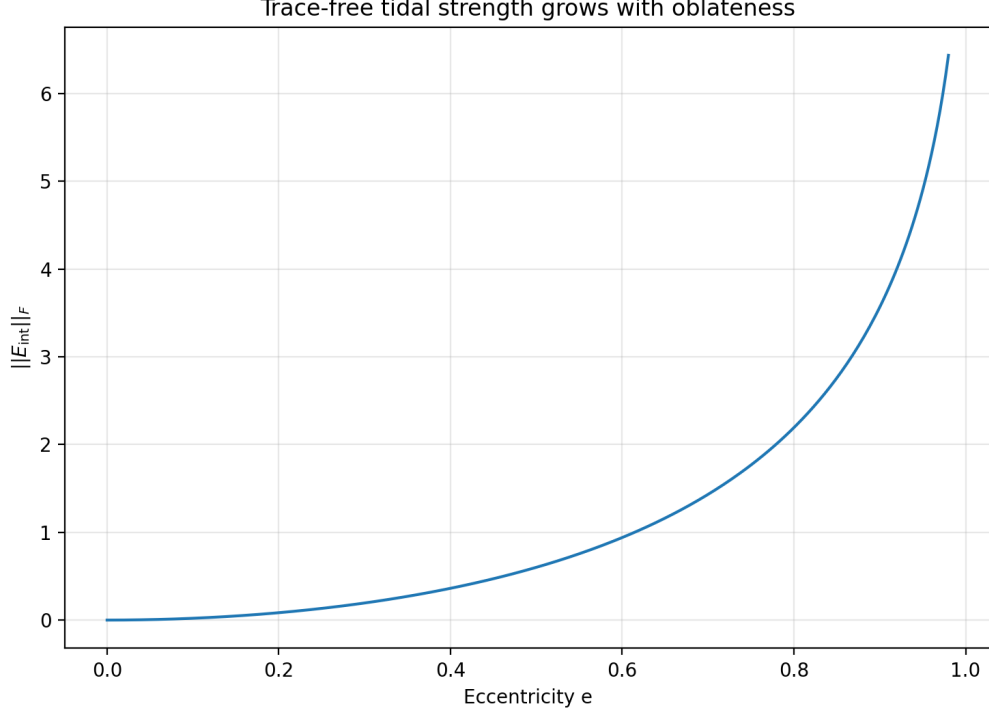


Figure 7: Analytic trace-free tidal strength versus eccentricity. The diagnostic vanishes in the spherical limit and grows monotonically with oblateness for the uniform-density oblate spheroid family.

13.7 Multipole compression witness

Exterior multipole compression was tested for the same fiducial spheroid. The axial multipole ratios include

$$q_2/q_0 = -0.15, \quad (70)$$

$$q_4/q_0 \approx 0.0482142857, \quad (71)$$

$$q_6/q_0 \approx -0.0200892857, \quad (72)$$

$$q_8/q_0 \approx 0.0095880682, \quad (73)$$

$$q_{10}/q_0 \approx -0.0049784200. \quad (74)$$

For a maximum angular relative potential error tolerance $\tau = 10^{-3}$, the compression-faithful radii are

$$L_{\text{max}} = 2 : \quad r_{\tau}/a \approx 2.61, \quad (75)$$

$$L_{\text{max}} = 4 : \quad r_{\tau}/a \approx 1.63, \quad (76)$$

$$L_{\text{max}} = 6 : \quad r_{\tau}/a \approx 1.33, \quad (77)$$

$$L_{\text{max}} = 8 : \quad r_{\tau}/a \approx 1.17, \quad (78)$$

$$L_{\text{max}} = 10 : \quad r_{\tau}/a \approx 1.08. \quad (79)$$

The monopole-only truncation did not reach $\tau = 10^{-3}$ by $r/a = 5$ in this scan. This confirms the expected compression behavior: increasing truncation order moves the compression-faithful radius inward. It remains an approximation-validity scale, not a parity surface.

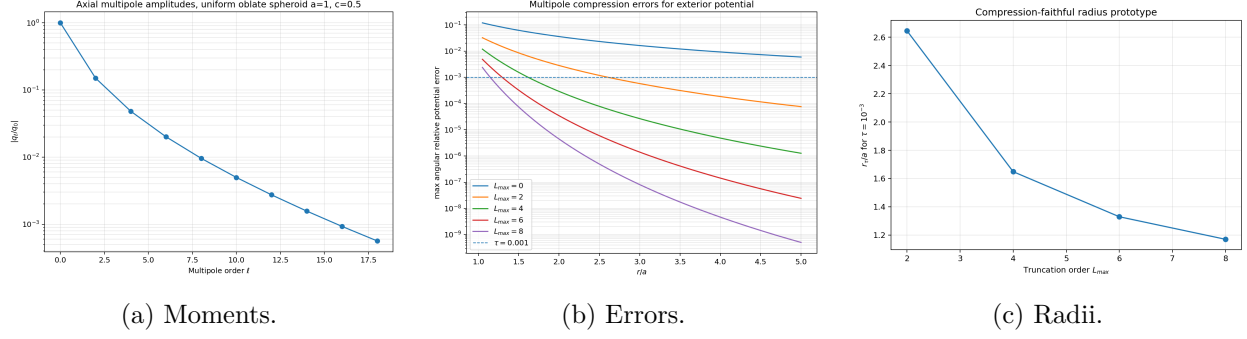


Figure 8: Exterior potential compression witnesses. Multipole amplitudes decrease with order, and higher truncation order reduces the radius at which a declared tolerance is met. For $\tau = 10^{-3}$, the quadrupole truncation reaches fidelity at $r_\tau/a \approx 2.61$, while $L_{\max} = 10$ reaches fidelity at $r_\tau/a \approx 1.08$. These radii are approximation-validity scales, not parity surfaces.

13.8 Scalar Ricci-compression comparison

The same fiducial spheroid was also evaluated with a traditional scalar finite-source readability branch, treating the spheroid as a discretized volume source with

$$Q(P) = \sum_s \frac{dm_s}{(|P - r_s|^2 + \epsilon^2)^{3/2}}. \quad (80)$$

This scalar diagnostic is not the Weyl-electric tensor. It is a compressed finite-source readability. Its role in this paper is comparative: does the scalar parity geometry duplicate the trace-free tidal attribution geometry, or does it expose a different bookkeeping channel?

For the north-south partition, the scalar method recovers the expected symmetry plane, as the Weyl-electric method does. For the asymmetric polar-cap-versus-belt partition with $\theta_c = \pi/4$, the scalar method behaves very differently. It produces an interior zero set, with sampled residual range

$$\Psi_{Q,\min} \approx -26.16, \quad \Psi_{Q,\max} \approx 22.88, \quad (81)$$

whereas the Weyl-electric Frobenius residual for the same partition remains strictly positive.

The volume-weighted diagnostic is especially suggestive. The cap occupies the mass fraction

$$M_{\text{cap}}/M_{\text{total}} = 1 - \cos(\pi/4) \approx 0.2929, \quad (82)$$

while the scalar positive-residual region occupies approximately

$$V_{\Psi_Q > 0}/V \approx 0.2839. \quad (83)$$

The match is not asserted as a theorem here. It is a prototype clue: the scalar parity geometry appears to shadow source-weight bookkeeping in a compressed Ricci-side channel.

The Weyl-electric method does not reproduce that volume partition. For the same cap-versus-belt split, the sampled Weyl-electric residual is positive throughout the interior. Its bookkeeping balance is not a volume equality but tensor closure:

$$E_{ij}^{(\text{cap})} + E_{ij}^{(\text{belt})} = E_{ij}^{(\text{total})}, \quad (84)$$

with maximum numerical closure error of order 10^{-13} in the prototype run. Thus the scalar and Weyl-electric channels agree where symmetry forces agreement, but diverge where the source

partition becomes anisotropic. This divergence is evidence that the scalar method is not simply a cruder drawing of the Weyl-electric geometry. It is a different compression readable.

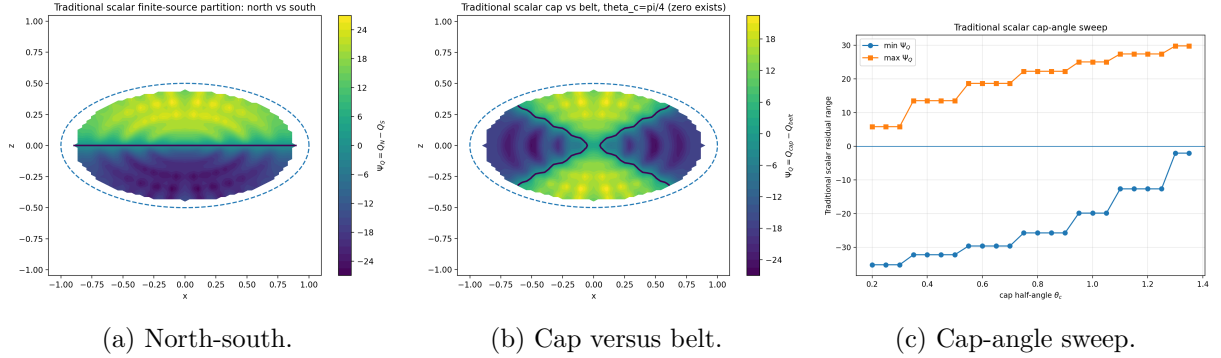


Figure 9: Traditional scalar finite-source comparison on the same fiducial spheroid. The scalar method recovers the expected north-south symmetry plane and produces cap-versus-belt parity geometry where the Weyl-electric Frobenius readable does not. This supports the interpretation that scalar parity is a compressed Ricci-side attribution readable, while Weyl-electric reconstruction preserves a higher-fidelity trace-free curvature channel.

14 Validation and reproducibility protocol

The minimum validation stack is: recover the analytic interior tensor; recover the Poisson trace; recover the analytic trace-free tensor; verify the spherical limit $E_{ij} = 0$; recover symmetry parity planes; report asymmetric partitions honestly as parity surfaces or residual witnesses; sweep eccentricity and cap angle; compute compression-faithful radii with declared tolerance and error norm; preserve all metadata.

The required metadata include body parameters, quadrature rule, boundary node count, observation grid, diagnostic readable, partition rule, error norm, tolerance, claim status, and figure status. The primary numerical object is a table of boundary-element tensor contributions, for example $dE_{ij}(r' \rightarrow P)$, indexed by boundary element, observation point, partition, diagnostic readable, and validation metadata. Figures are generated downstream from that table. They are witnesses only.

15 Limitations and extensions

The calculation is Newtonian. Full GR requires declaring $g_{\mu\nu}$, u^μ , $C_{\alpha\beta\gamma\delta}$, $E_{\mu\nu}$, matter model, foliation, gauge, boundary data, and initial data.

The body is uniform-density. In a polytrope or stratified body, $\nabla\rho$ has bulk support and the attribution record is not boundary-only. The scene is static. Rotation, magnetic Weyl, pressure, and frame dragging are deferred. The boundary is sharp. Smooth transition layers would smear boundary support into a shell.

The partitions are declared diagnostics. No partition is natural without symmetry, matter structure, eigenstructure, retention logic, or declared purpose. The compression radius depends on tolerance, truncation order, field quantity, angular norm, and origin. It is not a physical surface.

15.1 Speculative philosophical interpretation: curvature expression and the Ricci-Weyl handoff problem

The preceding sections contain the mathematical and numerical result: a static Newtonian weak-field reconstruction, a partitionable Weyl-electric attribution ledger, a scalar comparison, and declared compression witnesses. The present subsection is different in status. It is speculative philosophical interpretation. It does not add a theorem, modify GR, or claim a solved mechanism. Its purpose is to name the conceptual gap exposed by the calculation: how Ricci-side source custody becomes Weyl-electric expression dynamically, without appealing to an instantaneous update of the geometry.

The present construction is deliberately static. It reconstructs the trace-free tidal field of a declared source model and partitions the resulting attribution record. It therefore describes a solved weak-field geometry, not the dynamical process by which curvature expression propagates, hands off, or becomes collective.

This is not a small omission. It is the frontier exposed by the calculation. The static result shows that finite stress-energy structure can be registered as a Ricci-side matter account, that the same source geometry can be reconstructed through a Weyl-electric attribution ledger, and that exterior detail can be compressed into multipoles. What it does not yet explain is the mechanism by which these accounts transform into one another as a physical source changes.

In Newtonian potential theory, the field of a static source is obtained from an elliptic boundary-value problem. If read carelessly, that static solution can appear to update everywhere at once: change the source, solve the slice, and the distant field seems to have changed with no travel time. That apparent instant updating is a contradiction only if the static solution is mistaken for a dynamical mechanism. In general relativity, it cannot be the physical story. A deformation of the source, a rotation of its principal axes, a redistribution of stress-energy, or a change in boundary structure cannot be expressed outside the future light cone of the change. The geometry may be solved as a slice, but curvature expression must propagate as a causal process.

Accordingly, the attribution geometry described here should be understood as a frozen-slice reconstruction. It identifies how source structure is represented in a static weak-field solution, but it does not yet supply the transfer function from source-local Ricci custody to Weyl-electric field expression, nor from individual Weyl-electric custody to parent-field or collective multipole expression.

The missing object is a curvature-expression handoff mechanism: a dynamical account of how stress-energy-supported Ricci structure is converted into, coupled to, or expressed through trace-free Weyl structure; how changes in that structure propagate at causal speed; how neighboring finite sources reorganize the resulting tensor field between them; and how individual source attribution compresses into collective exterior geometry. The open question is not whether GR permits causal propagation. It does. The open question is how to read, ledger, and visualize the causal transformation of curvature custody inside the standard theory. The static spheroid ledger shows that the accounts can be separated and closed on one slice. It does not show the causal movie by which the accounts transform.

This is not a proposed modification of gravity. It is an open readability problem inside standard gravity. The static result supplies the ledger. The unresolved dynamical problem is how that ledger changes, propagates, and becomes collective across scale without appealing to instantaneous updating, and without treating the solved geometry itself as the mechanism. This is why the handoff problem is philosophical in motivation but technical in demand: the words *custody*, *expression*, and *handoff* must eventually cash out as causal, tensor-level dynamics rather than attractive metaphor.

Natural extensions include triaxial ellipsoids, polytropes, slow-rotation post-Newtonian bodies, Schwarzschild interior/exterior tests, full relativistic rotating-star models, comparison against tidal

eigenstructure witnesses, and omitted-multipole parity against environmental context.

16 Conclusion

This paper developed a single-body attribution diagnostic for the trace-free tidal field of a uniform-density oblate spheroid. Its technical result is a boundary-supported reconstruction of the interior tidal tensor and a partitionable attribution ledger for the Weyl-electric weak-field proxy.

The larger lesson is sharper. Finite matter expresses a finite curvature budget inside the continuum model of gravity. The field does not end at the matter boundary, and no wall is introduced. But curvature custody is not featureless. Ricci-side bookkeeping records where mass-energy is locally accounted for as curvature. Weyl-electric bookkeeping records how source shape and anisotropy are expressed as trace-free tidal geometry. Multipole compression records when resolved detail may be replaced, for a declared tolerance, by a lower-order exterior representation.

In the fiducial spheroid $a = 1, c = 0.5, \rho_0 = G = 1$, the boundary-supported reconstruction recovers the analytic interior tidal tensor to near machine precision on compact interior points. Symmetry partitions recover the expected parity plane. The $\theta_c = \pi/4$ polar-cap partition produces no interior equality surface for the Weyl-electric Frobenius readable, and is therefore correctly classified as a residual-foliation witness. Exterior multipole truncation produces compression-faithful radii that move inward with increasing $L_{\max}$.

The scalar comparison adds the decisive interpretive clue. A traditional finite-source scalar readability applied to the same cap-versus-belt partition produces parity geometry whose volume fractions closely shadow the source partition. The Weyl-electric method does not reproduce that volume split. Instead, it preserves tensor closure to numerical precision while producing no equality surface. The two methods are not duplicates. They are different readings of the same finite source: scalar parity as compressed Ricci-side attribution, and Weyl-electric attribution as trace-free curvature custody.

The construction does not make Weyl curvature local, material, or newly sourced. It makes one weak-field Weyl-electric proxy auditable in one declared single-body model. The source remains one body. Boundary elements are derived support, not new matter. Residual contours are not parity surfaces unless equality exists. Compression radii are approximation-validity scales unless a separate equality rule is declared.

The present result is therefore a static attribution theorem-in-practice, not a full handoff theory. It shows that finite-source trace-free tidal structure can be reconstructed, partitioned, compared, and closed in a solved weak-field geometry. It does not yet explain the deeper mechanism by which Ricci-side custody becomes Weyl-electric expression, how a change in that expression propagates from a source, how neighboring finite bodies reorganize curvature custody between them, or how local attribution becomes collective exterior geometry. Those questions require dynamics.

The speculative philosophical interpretation exposed by the calculation is this: the field is seamless, but curvature custody changes hands. In the continuum description used here, finite stress-energy accounts can be locally registered, trace-free structure can be carried outward, and resolved detail can be compressed into context. But the geometry itself is not yet explained as a causal transformation. A static solution shows the before-and-after ledger; it does not show how the ledger updates without violating causality. If source deformation altered distant Weyl-electric structure instantly, that reading would contradict the causal character of GR. The instant-updating picture is therefore not an option; it is the sign that the static ledger is missing its dynamical handoff mechanism. The open question is the lawful causal process by which Ricci-supported source structure becomes changing Weyl-electric geometry, and by which that geometry is handed into

context.

A finite body does not need a hard edge in the field to have readable curvature identity. Its matter support, Ricci-side account, Weyl-electric expression, and exterior compression can be separated as bookkeeping channels inside one continuous geometry. On the far side of a source-local attribution boundary, the field does not go blank and the source does not vanish; rather, curvature custody is handed into context. The static ledger shown here does not yet explain the living mechanism of that handoff. It names the problem sharply enough to pursue it.

The claim is scoped to the continuum model used here. It does not settle whether spacetime is ultimately continuous or discrete at a deeper quantum-gravitational level. The result is more modest and more useful: within the declared GR/weak-field regime, finite matter can carry a finite, computable curvature budget whose expression is structured even though the field remains smooth. The next problem is not another static surface. It is the causal geometry of curvature expression: the worldtube through which source-local Ricci custody becomes Weyl-electric field structure, and through which individual custody becomes collective context.

A Model declaration

Model: Uniform-density oblate spheroid.

Purpose: Test whether a single extended Newtonian source admits boundary-supported attribution residuals for trace-free tidal structure.

Regime: Newtonian weak-field limit with static electric-Weyl interpretation.

Coordinates: Body-centered Cartesian coordinates (x, y, z) , symmetry axis z .

Domain: Interior Ω , boundary $\partial\Omega$, and exterior weak-field region.

Physical source: One uniform-density spheroid.

Derived supports: Boundary quadrature elements representing the density-gradient support at $\partial\Omega$.

Interrogation points: Interior observation points, meridional grid points, exterior radial shells.

Field quantities: Newtonian potential Φ , tidal tensor T_{ij} , trace-free tidal tensor E_{ij} .

Diagnostics: Tensor E_{ij} , Frobenius norm, signed projection, eigenstructure, exterior multipole truncation error.

Validation: Analytic ellipsoid tensor, Poisson trace, spherical limit, symmetry partition planes, compression-radius convergence.

Figures: Explanatory witnesses only.

B Boundary parameterization and implementation form

For numerical reconstruction, parameterize the spheroid boundary by

$$r'(\theta, \phi) = (a \sin \theta \cos \phi, a \sin \theta \sin \phi, c \cos \theta), \quad (85)$$

with $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$. The outward vector area element is

$$\hat{n} dS = \left(ac \sin^2 \theta \cos \phi, ac \sin^2 \theta \sin \phi, a^2 \sin \theta \cos \theta \right) d\theta d\phi. \quad (86)$$

Using the vector area element $dS_i = \hat{n}_i dS$, the symmetrized boundary formula becomes

$$T_{ij}(P) = -\frac{G\rho_0}{2} \int \frac{dS_i R_j + dS_j R_i}{R^3}. \quad (87)$$

This avoids separately computing $\hat{n}$ and dS , and it reduces antisymmetric quadrature noise.

For implementation, each boundary quadrature element should store $r'(\theta, \phi)$, $d\vec{S}(\theta, \phi)$, and the quadrature weight, or the equivalent quadrature-weighted vector area element. For each observation point P , compute $R = P - r'$ and $R = |R|$. Then accumulate

$$dT_{ij} = -\frac{G\rho_0}{2} \frac{dS_i R_j + dS_j R_i}{R^3}, \quad (88)$$

and

$$dE_{ij} = dT_{ij} - \frac{1}{3} \delta_{ij} dT^k_k. \quad (89)$$

The final tensor is $T_{ij}(P) = \sum_{\partial\Omega} dT_{ij}$ and $E_{ij}(P) = \sum_{\partial\Omega} dE_{ij}$.

C Numerical data products

Minimum contribution table fields include model identifier, spheroid parameters a, c, ρ_0, G , eccentricity, boundary node identifier, boundary position, boundary vector area, boundary weight, observation identifier, observation position, the independent components of dT_{ij} and dE_{ij} , partition identifier, partition label, diagnostic identifier, claim status, and run identifier.

Suggested sidecar tables include a model table, boundary table, observation table, partition table, diagnostic table, validation table, compression table, and figure manifest.

D Figure protocol

Figure 1 would show the declared model and boundary support, with schematic status. Figure 2 is boundary reconstruction convergence, with validation status. Figure 3 is boundary attribution density, with witness-map status. Figure 4 is symmetry partition parity, with zero-set validation status. Figure 5 is polar-cap residual foliation, with asymmetric-witness status unless $\Psi = 0$ exists. Figure 6 is eccentricity scaling of $\|E_{\text{int}}\|_F$, with analytic scaling status. Figure 7 is multipole compression, with compression-witness status, not parity-surface status.

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